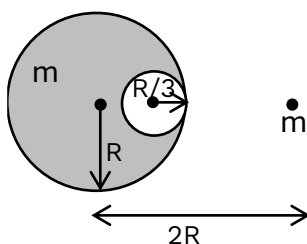
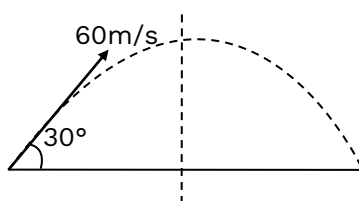


PHYSICS

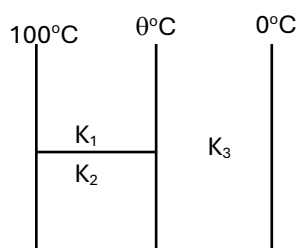
- Q1** Write dimensional formula of $\frac{B}{\mu_0}$. Where B = magnetic field and μ_0 = magnetic permeability.
- Q2** Find the ratio of gravitational force $\frac{F_1}{F_2}$ between sphere and mass particle before cutting and after cutting.



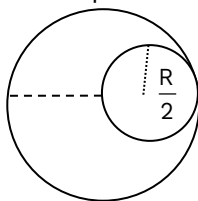
- Q3** Two soap bubbles of radius $r_1 = 3$ cm and $r_2 = 5$ cm combine. Find radius of curvature of the common surface separating them
- Q4** 10^{-3} kg ice at -10°C is converted into steam at 110°C find heat required in this process.
- Q5** A closed organ pipe and an open organ pipe are in resonance with a same tuning fork, the closed organ pipe resonates in 9th harmonic and the open organ pipe resonance in 4th harmonic if the length of closed organ pipe is 10 cm, the length of the open organ pipe is:
- Q6** Find the ratio of heights in first and last second of upward journey



- Q7** Find out the temperature ' θ ' shown in diagram, $K_1 = 60$, $K_2 = 120$ and $K_3 = 135$

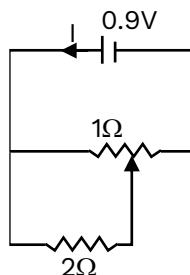


- Q8** From a sphere of mass M and radius R , a cavity of radius $\frac{R}{2}$ is created. Find the moment of inertia about an axis passing through the centre of sphere and cavity



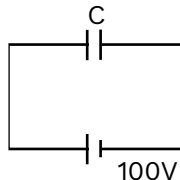
- (1) $\frac{31}{48}MR^2$ (2) $\frac{31}{80}MR^2$ (3) $\frac{13}{32}MR^2$ (4) $\frac{21}{32}MR^2$

- Q9** Jokey is connected at mid-point. Find current through battery.

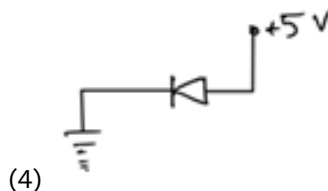
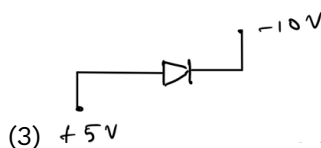
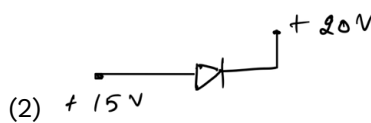
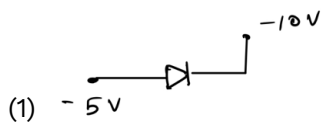


- Q10** The ground state radius of Hydrogen atom is a_0 . Calculate the radius of first excited state of He^+ ion.

- Q11** A dielectric is inserted $k = 2$. Find ΔU (change in energy stored in capacitor), Δq (change in charge on capacitor).



- Q12** Which of the following is forward biased?



- Q13** An electron in the ground state of the hydrogen atom has the orbit radius of $5.3 \times 10^{-11} \text{ m}$, while that for the electron in third excited state is $8.48 \times 10^{-10} \text{ m}$. The ratio of the de-Broglie wavelength of electron in the ground state to that in the excited state is
- (1) 9 (2) 3 (3) 4 (4) 16

- Q14** The energy radiated by the sphere of radius 0.2 m and surface temperature 800 K is E . Then the energy radiated by the sphere of radius 0.8 m and surface temperature 400 K is:

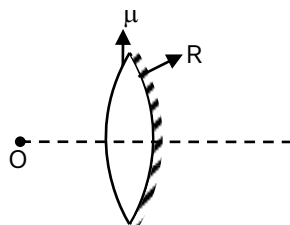
Q15 Assertion:- The fringe width in the YDSE decreases if the setup is put into a denser medium.

Reason:- Velocity of light decreases in the denser medium but frequency remains constant.

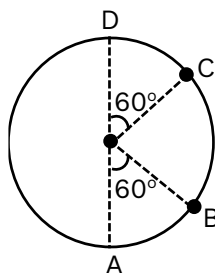
Q16 Assertion:- In a vernier callipers, one vernier scale division is smaller than one main scale division.

Reason:- The vernier constant is given by one main scale division multiplied by the number of vernier scale divisions.

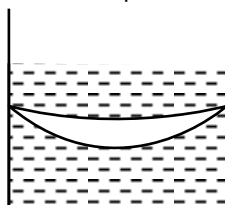
Q17 Object and image coincide at given point O. Find object distance.



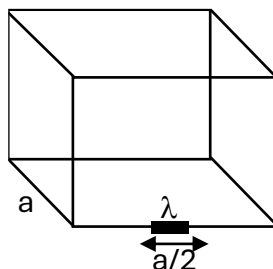
Q18 A particle is projected with a velocity v_0 from point A in the vertical circular path, such that the rope becomes slack at point D, then the ratio of kinetic energy at point B and C $\left(\frac{KE_B}{KE_C}\right)$.



Q19 A concavo-convex lens of refractive index $\frac{3}{2}$ is placed in water $\left(\mu = \frac{4}{3}\right)$. If the radius of curvature of inner and outer surface are R_1 and R_2 . Find the power of combination.



Q20 A line charge of length $a/2$ and charge density λ is placed along the edge of the cube. Find the flux through cube.



Q21 A light of wavelength 525 nm is incident on cesium ($\phi = 1.9$ eV) and lithium ($\phi = 2.5$ eV). Then the electron is emitted from which element?